AKARSHA JAIN
Los Angeles | akarshajain888@gmail.com |+1 213 301 3453 | LinkedIn| Github

EDUCATION

University of Southern California , *Los Angeles, USA* — Masters in Computer Science — **Jan, 2025 – Dec, 2026**
Walchand Institute of Technology, *Solapur, India* — Bachelor of Technology In Computer Science — **Aug, 2019 – Jun, 2023**

TECHNICAL SKILLS

Programming Languages: Java, Python, C++, Kotlin, C, SQL
Software Development: Object-Oriented Programming (OOP), Data Structures & Algorithms, RESTful API Design, Microservices Architecture, Android MVVM Architecture, Front-End Development (React.js,Three js, HTML, CSS)
Frameworks & Tools: Spring Boot, Hibernate, Maven, Gradle, Flask, Node.js, Three.js, Retrofit, OkHttp
Cloud Platforms: AWS (EKS, ECS, EC2, S3, RDS, EFS) **| Containerization & Orchestration:** Docker, Rancher, Helm
DevOps & CI/CD: Jenkins, GitHub Actions, Harness, Build Automation (Maven, Gradle)
Machine Learning & Computer Vision: TensorFlow, YOLOv8, OpenCV, Scikit-learn, NumPy, Pandas, Matplotlib
Databases: PostgreSQL, MySQL, Flyway (Schema Migration)
Testing & Security: JUnit, Mockito, SonarQube, OAuth2, JWT **| Version Control:** Git, GitHub, GitLab, Bitbucket.

WORK EXPERIENCE

USC Summer Program (USC Viterbi K–12 Program)| *Teaching Assistant* | *Los Angeles, California, United States* **May 2025-Present**

- Taught and guided **60+ students** in **programming and robotics** through hands-on activities using autonomous EZ-Bots, which were coded to **detect human presence** and safely navigate around them, ensuring a **real-world application** of **obstacle avoidance algorithms**.
- **Delivered lectures** and led **Zumi and Red Bot robotics workshops**, programming autonomous bots with pathfinding, **ML-based obstacle detection**, and line-following logic using calibrated sensors and real-time feedback loops.
- **Conducted Python programming sessions** to program **thermal-sensor drones**, integrating sensor data streams from **AirBeam3** and **Databot kits** to simulate **environmental sensing** and **data-driven decision making**.

National Australia Bank (NAB Innovation Center India) | *Software Engineer* | *Gurugram, Haryana, India* **July 2023-December 2024**

- Engineered and maintained **6+ Java/Spring Boot microservices** processing **1M+ daily transactions** with **<100ms latency**, **ensuring real-time fraud detection** responsiveness.
- Reduced transaction latency **by 40%** through Redis caching, JDBC connection pool tuning, and SQL query optimization on **5M+ records**.
- Designed and implemented real-time alerting pipelines using Apache Kafka streams, improving fraud detection accuracy and reducing false positives by **25%.**
- Developed and secured 3D Secure (3DS) authentication APIs with **OAuth2** and **JWT**, enabling **multi-wallet (Apple/Samsung Pay) fraud reconciliation.**
- Containerized microservices with **Docker** and automated zero-downtime **Helm** deployments on **AWS EKS**, sustaining **99.99% uptime**; resolved **JVM warm-up** latency issues, earning **Spot Award**.

Persistent | *Martian Program PSPL Intern* | *Pune, Maharashtra, India* **May 2022-June 2022**

- Developed a single-page **React.js** application with dynamic routing powered by React Router, ensuring smooth navigation and fast page loads.
- Built **RESTful APIs** using **Node.js** and **Express.js** backend, implementing **JWT-based authentication** with secure token storage and refresh logic.
- Implemented persistent authentication state across sessions using localStorage and context API, preventing user logout on page reloads.
- Conducted **cross-browser testing** and **debugging** to ensure consistent user experience across Chrome, Firefox, and Safari.

Blue Planet Infosolution | *Android App Development Intern* | *Remote, India* **November 2021-January 2022**

- Architected and implemented modular Android applications using **Kotlin** and **Java** with **MVVM architecture,** ensuring maintainability and testability.
- Integrated **RESTful APIs** with **Retrofit** and **OkHttp** to enable seamless backend communication, implementing robust error handling and offline caching.
- Improved Android app stability **by 40%** through **lifecycle optimization**, memory leak detection (**LeakCanary**), async handling with Coroutines, **UI enhancements** with Material Design, performance tuning via Android Profiler, and robust test coverage using **Espresso** and **JUnit.**

PROJECTS

Smart Glasses | Incubated under TATA Technologies

- Engineered smart glasses for visually impaired users using Raspberry Pi 5, camera module, **YOLOv8**, and **CNNs** for real-time object detection, with precise classification via **OpenCV, TensorFlow**, and non-max suppression. ***Winner, TATA MAGIC Innovation Challenge.***
- **Integrated NLP** and **Google TTS** to convert detected visual objects into spoken feedback through in-ear audio; optimized for low power consumption and scalability with **facial recognition and GPS navigation**.

Toy car Automation using ML

- Built an autonomous toy car with Raspberry Pi**, YOLOv8**, camera, sensors for real-time object detection, obstacle avoidance, and road navigation.
- Integrated computer vision and **GPIO-based motor** control to enable intelligent path planning, showcasing applied ML, CV, and embedded systems skills.

SkillBridge – AI-Enhanced Skill Tracking Platform

- Designed and developed a full-stack learning tracker platform to help users manage and to track learning goals with secure **RESTful APIs** (Spring Boot), **JWT authentication**, and **PostgreSQL integration**.
- Integrated **OpenAI API** for personalized learning suggestions and deployed via **Docker** on **AWS** with CI/CD using GitHub Actions.

LEADERSHIP & COMMUNITY ENGAGEMENT

- Graduate Student Affairs Officer, **Society of Women Engineers (SWE) @ USC |** Founder & Lead, **Coding Club**, Walchand Institute of Technology.
- Awarded funding and won in the Magic competition with team Electra, in collaboration with **TATA Technologies Ltd**.
- **Received Spot Award** recognition on a quarterly basis for exceptional team performance and critical production issue resolution.